THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 3 & 4: **A Story of Numbers, Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 149	Add the values, subtracting a smaller symbol that comes before a larger one
2	**(a)** XXXVIII	38 = XXXVIII
3	**(b)** 1101	Repeated division of 13 by 2
4	**(d)** 222	Repeated division of 26 by 3
5	**(d)** 13	Add digit × place value in base 2
6	**(d)** 95°	Angle sum of a quadrilateral is 360°
7	**(d)** 115°	Adjacent angles are supplementary
8	**(b)** 5 cm	Diagonals bisect at right angles; use Pythagoras
9	**(c)** 10 cm	$\sqrt{l^2 + b^2}$
10	**(a)** trapezium	Trapezium

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Al-Khwarizmi and Al-Kindi used the name Hindu numerals (1); Fibonacci promoted them in Europe (1).	1+1
12	Counting by marking a stroke for each object, grouped in fives (2).	2
13	No, the sum would be less than 360° (2).	2
14	180° (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Expand using powers of 2 (2); value $= 27$ (1).	2+1
16	Area $= \frac{1}{2} d_1 d_2 = \frac{1}{2} \times 12 \times 16$ (2); 96 cm² (1).	2+1
17	Expand using powers of 3 (2); value $= 73$ (1).	2+1
	OR: Expand using powers of 5 (2); value $= 194$ (1).	2+1
18	Both diagonals are hypotenuses of right triangles with legs l and b (2); so both $= \sqrt{l^2 + b^2}$ (1).	2+1
	OR: $\frac{1}{2}(10 + 6) \times 4$ (2); 32 cm² (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Maya base 20, Mesopotamian base 60; Maya had a shell for zero (3). (ii) Both used place value (2).	3+2
	OR: (i) An ancient Indian manuscript from about the 3rd century CE (2). (ii) It shows a dot used as zero (3).	2+3
20	(i) $D = 110°$, $C = 100°$ (3). (ii) Co-interior angles are supplementary (2).	3+2
	OR: (i) Adjacent angle bisectors meet at 90° because half of the supplementary angles is 90° (3). (ii) 90° (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 3 & 4: **A Story of Numbers, Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 23	Add digit × place value in base 5
2	**(d)** Aryabhata	Aryabhatiya
3	**(a)** 20	Vigesimal
4	**(d)** 2	Digits are 0 and 1
5	**(c)** sahasra	Thousand
6	**(c)** parallel and equal	Basic property
7	**(a)** parallelogram	Opposite sides are parallel
8	**(b)** all sides are equal	Rhombus
9	**(a)** square	Both properties
10	**(c)** A and C	Diagonals bisect vertex angles

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$40 + 4 = 44$ (2).	2
12	60 minutes (1); our time measurement still uses 60 (1).	1+1
13	Rhombus, square and kite (2).	2
14	AC = BD = 26 cm (1); they bisect at the centre, each half 13 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Expand using powers of 8 (2); value $= 455$ (1).	2+1
16	Only the square (1.5); because a rhombus has unequal diagonals (1.5).	1½+1½
17	Hindu uses place value and zero (1.5); Roman has no zero and needs many symbols (1.5).	1½+1½
	OR: Used base 60 (1); two wedge symbols for 1 and 10 (1); place value without a true zero in early times (1).	1+1+1
18	Parallelogram, rectangle, rhombus, square (2); property of parallelograms (1).	2+1
	OR: Angle sum $= 360^\circ$ (1); $C = 360 - 90 - 90 - 70$ (1); 110° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 0.1_2 and 1001011_2 (3). (ii) $2^3 = 8$ (2).	3+2
	OR: (i) Base 5 (1). (ii) $163 = 1 \times 125 + 1 \times 25 + 2 \times 5 + 3$ i.e. 1123_5 (3). (iii) 1, 1, 2, 3 (1).	1+3+1
20	(i) Half diagonals 12 and 5; other diagonal 10 cm (2). (ii) $\frac{1}{2} \times 24 \times 10 = 120$ cm² (1). (iii) $30000/120 = 250$ tiles (2).	2+1+2
	OR: (i) $36^\circ, 72^\circ, 108^\circ, 144^\circ$ (3). (ii) Yes, if the sides with 36° and 144° are parallel; angles sum to 180° (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **A Story of Numbers, Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 444	Add the values, subtracting a smaller symbol that comes before a larger one
2	**(c)** LXXVI	76 = LXXVI
3	**(c)** 11001	Repeated division of 25 by 2
4	**(d)** 123	Repeated division of 38 by 5
5	**(b)** 22	Add digit × place value in base 2
6	**(c)** 100°	Angle sum of a quadrilateral is 360°
7	**(b)** 108°	Adjacent angles are supplementary
8	**(c)** 13 cm	Diagonals bisect at right angles; use Pythagoras
9	**(a)** 13 cm	$\sqrt{l^2 + b^2}$
10	**(b)** equal and bisect each other	Rectangle property

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$7 \times 64,\ 3 \times 8,\ 4 \times 1$ (2).	2
12	0 and 1 (1); 101_2 (1).	1+1
13	A rectangle (2).	2
14	$180 - 65 = 115°$ each (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	From India to the Arab world around 800 CE (1.5); then to Europe via Fibonacci and others around 1100–1200 CE (1.5).	1½+1½
16	$\sqrt{40^2 + 30^2}$ (2); 50 m (1).	2+1
17	Separate symbols for 1, 10, 100, 1000 and so on (1.5); repeated to show quantities, additive (1.5).	1½+1½
	OR: $1440 + 560 = 2000$ (2); MM (1).	2+1
18	Perimeter $= 2(12 + 9) = 42$ cm (1.5); area needs the height, not just sides (1.5).	1½+1½
	OR: Sum: $4x + 120 = 360$ (1.5); $x = 60$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) No place value (2). (ii) $9 \times 11 = 99$ is XCIX, but the method is not systematic (3).	2+3
	OR: (i) $1,00,000$ and $1,00,00,000$ (3). (ii) 100 (2).	3+2
20	(i) A diagonal gives congruent triangles, so alternate angles equal (3). (ii) A folding gate or a parallelogram linkage (2).	3+2
	OR: (i) Side $= 10$ cm (3). (ii) 100 cm² (2).	3+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 3 & 4: **A Story of Numbers, Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 29	Add digit × place value in base 8
2	**(c)** Al-Khwarizmi	He wrote on calculation with Hindu numerals
3	**(a)** D	I=1, V=5, X=10, L=50, C=100, D=500, M=1000
4	**(c)** 0, 1, 2, 3, 4	Digits are less than the base
5	**(b)** 800 CE	Later reached Europe
6	**(b)** kite	Kite
7	**(a)** rectangle	Rectangle property
8	**(d)** 110°	Opposite angles are equal
9	**(b)** kite (in general)	Kite need not be parallel
10	**(d)** parallelogram	Converse property

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They learnt them from the Arab world (2).	2
12	It holds a place when a position is empty, distinguishing 305 from 35 (2).	2
13	$x + 4x = 180$, $x = 36^\circ$ and 144° (2).	2
14	It has all sides equal and also right angles (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	I, II, III, IV, V, VI, VII, VIII, IX, X (3).	3
16	Only one pair of opposite sides is parallel in a trapezium (3).	3
17	In base b digits run from 0 to $b-1$ (1.5); for $b=2$ these are 0 and 1 (1.5).	1½+1½
	OR: $37 = 6\times6+1$ (2); 101_6 (1).	2+1
18	A rhombus (1); each side is half a diagonal and diagonals are equal (2).	1+2
	OR: $\frac{1}{2}\times14^2 = 98$ cm² (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) India, Arab world around 800 CE, Europe around 1100–1200 CE (3). (ii) Al-Khwarizmi and Fibonacci (2).	3+2
	OR: (i) $1, 2, 10, 11, 12, 20, 21, 22, 100, 101$ (3). (ii) 3 and higher digits (2).	3+2
20	(i) $\frac{1}{2}(90+60)\times40 = 3000$ m² (2). (ii) ₹15000 (1). (iii) One pair of sides parallel (2).	2+1+2
	OR: (i) Angle inside at a vertex; exterior 90° (2). (ii) Each exterior = 180° − interior; total $720-360=360^\circ$ (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.